

BDA Practical No-5 using Ubuntu

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Step 1:-Open Ubuntu Terminal and type the command given below to verify whether Mongo DB is installed or not on the system

mongod --version

If Mongo DB is installed then the output will show you the version number of Mongo DB like shown below:-

```
aditya@DESKTOP-4CC1PF9:~$ mongod --version
db version v8.0.21
Build Info: {
  "version": "8.0.21",
  "gitVersion": "e67138ffda9f90b59531128bc43678bb5ad105ed",
  "opensslVersion": "OpenSSL 3.0.13 30 Jan 2024",
  "modules": [],
  "allocator": "tcmalloc-google",
  "environment": {
    "distmod": "ubuntu2404",
    "distarch": "x86_64",
    "target_arch": "x86_64"
  }
}
```

If Mongo DB is not installed then refer my “Mongo DB Installation for Ubuntu”pdf to install Mongo DB

Step 2:-Now type the following command:-

sudo service mongod start

```
aditya@DESKTOP-4CC1PF9:~$ sudo service mongod start
```

Step 3:-Now type:-

sudo service mongod status

You will see the output like shown below. You will see green text saying Active: active (running) means Mongo DB is running actively:-

```
aditya@DESKTOP-4CC1PF9:~$ sudo service mongod status
● mongod.service - MongoDB Database Server
   Loaded: loaded (/usr/lib/systemd/system/mongod.service; disabled; preset: enabled)
   Active: active (running) since Sun 2026-05-10 03:38:58 UTC; 1min 3s ago
     Docs: https://docs.mongodb.org/manual
    Main PID: 1405 (mongod)
  Memory: 291.3M (peak: 373.3M)
    CPU: 1.775s
   CGroup: /system.slice/mongod.service
           └─1405 /usr/bin/mongod --config /etc/mongod.conf

May 10 03:38:58 DESKTOP-4CC1PF9 systemd[1]: Started mongod.service - MongoDB Database Server.
May 10 03:38:59 DESKTOP-4CC1PF9 mongod[1405]: {"t":{"$date":"2026-05-10T03:38:59.976Z"},"s":"I", "c":"CONTROL", "id":
lines 1-12/12 (END)
```

Now to exit press “q”

Step 4:-Now type the following command:-

mongosh

You will output like shown below. Now You can type your Mongo DB commands:-

```
aditya@DESKTOP-4CC1PF9:~$ mongosh
Current Mongosh Log ID: 69fffd0789e74b80e9df8a2
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2
.8.3
Using MongoDB:      8.0.21
Using Mongosh:      2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2026-05-10T03:39:00.351+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See
http://dochub.mongodb.org/core/prodnotes-filesystem
2026-05-10T03:39:01.167+00:00: Access control is not enabled for the database. Read and write access to data and conf
figuration is unrestricted
2026-05-10T03:39:01.168+00:00: For customers running the current memory allocator, we suggest changing the contents o
f the following sysfsFile
2026-05-10T03:39:01.168+00:00: For customers running the current memory allocator, we suggest changing the contents o
f the following sysfsFile
2026-05-10T03:39:01.168+00:00: We suggest setting the contents of sysfsFile to 0.
2026-05-10T03:39:01.168+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----

test> |
```

Step 5:-Now type the following command:-

use studentDB

```
test> use studentDB
switched to db studentDB
```

You can have any other name as well instead of “studentDB” for database

Step 6:-Now type:-

db.createCollection("students")

```
studentDB> db.createCollection("students")
{ ok: 1 }
```

You can have any other name as well instead of “students” for collection

Step 7:-Now type the following command:-

db.students.insertOne({rollno: 1, name: "Rutvik", marks: 80, city: "Pune"})

```
studentDB> db.students.insertOne({rollno: 1, name: "Rutvik", marks: 80, city: "Pune"},
{
  acknowledged: true,
  insertedId: ObjectId('6a0012e5789e74b80e9df8b6')
})
```

Step 8:-Now type:-

```
db.students.insertMany([
  {rollno: 2, name: "Aditya", marks: 85, city: "Nashik"},
  {rollno: 3, name: "Priya", marks: 90, city: "Nashik"},
  {rollno: 4, name: "Amit", marks: 75, city: "Mumbai"},
  {rollno: 5, name: "Sneha", marks: 82, status: "pass", city: "Mumbai"}
])
```

```
studentDB> db.students.insertMany([
|   {rollno: 2, name: "Aditya", marks: 85, city: "Nashik"},
|   {rollno: 3, name: "Priya", marks: 90, city: "Nashik"},
|   {rollno: 4, name: "Amit", marks: 75, city: "Mumbai"},
|   {rollno: 5, name: "Sneha", marks: 82, status: "pass", city: "Mumbai"}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a0012f1789e74b80e9df8b7'),
    '1': ObjectId('6a0012f1789e74b80e9df8b8'),
    '2': ObjectId('6a0012f1789e74b80e9df8b9'),
    '3': ObjectId('6a0012f1789e74b80e9df8ba')
  }
}
```

Step 9:-Now type:-

```
db.students.find().pretty()
```

```

L
{
  _id: ObjectId('6a0012e5789e74b80e9df8b6'),
  rollno: 1,
  name: 'Rutvik',
  marks: 80,
  city: 'Pune'
},
{
  _id: ObjectId('6a0012f1789e74b80e9df8b7'),
  rollno: 2,
  name: 'Aditya',
  marks: 85,
  city: 'Nashik'
},
{
  _id: ObjectId('6a0012f1789e74b80e9df8b8'),
  rollno: 3,
  name: 'Priya',
  marks: 90,
  city: 'Nashik'
},
{
  _id: ObjectId('6a0012f1789e74b80e9df8b9'),
  rollno: 4,
  name: 'Amit',
  marks: 75,
  city: 'Mumbai'
},
{
}

```

Step 10:-Now type to print the students who marks are greater than 80:-

```
db.students.find({marks: {$gt: 80}}).pretty()
```

```

studentDB> db.students.find({marks: {$gt: 80}}).pretty()
[
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b7'),
    rollno: 2,
    name: 'Aditya',
    marks: 85,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b8'),
    rollno: 3,
    name: 'Priya',
    marks: 90,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8ba'),
    rollno: 5,
    name: 'Sneha',
    marks: 82,
    status: 'pass',
    city: 'Mumbai'
  }
]

```

Step 11:-Find any student who lives in either Nashik OR Mumbai.

```
db.students.find({city: {$in: ["Nashik", "Mumbai"]}}).pretty()
```

```

studentDB> db.students.find({city: {$in: ["Nashik", "Mumbai"]}}).pretty()
[
  {
    _id: ObjectId('6a001d62789e74b80e9df8c1'),
    rollno: 2,
    name: 'Aditya',
    marks: 85,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a001d62789e74b80e9df8c2'),
    rollno: 3,
    name: 'Priya',
    marks: 90,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a001d62789e74b80e9df8c3'),
    rollno: 4,
    name: 'Amit',
    marks: 75,
    city: 'Mumbai'
  },
  {
    _id: ObjectId('6a001d62789e74b80e9df8c4'),
    rollno: 5,
    name: 'Sneha',
    marks: 82,
    status: 'pass',
    city: 'Mumbai'
  }
]

```

Step 12:- Update Amit's marks to 80 using following command:-

```
db.students.updateOne({rollno: 3}, {$set: {marks: 80}})
```

```

studentDB> db.students.updateOne({rollno: 3}, {$set: {marks: 80}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

Step 13:- If we want to find everyone but sort them by marks (Highest to Lowest). The '-1' means descending order. '1' would be ascending.

```
db.students.find().sort({marks: -1}).pretty()
```

```
[
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b7'),
    rollno: 2,
    name: 'Aditya',
    marks: 85,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8ba'),
    rollno: 5,
    name: 'Sneha',
    marks: 82,
    status: 'pass',
    city: 'Mumbai'
  },
  {
    _id: ObjectId('6a0012e5789e74b80e9df8b6'),
    rollno: 1,
    name: 'Rutvik',
    marks: 80,
    city: 'Pune'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b8'),
    rollno: 3,
    name: 'Priya',
    marks: 80,
    city: 'Nashik'
  },
]
```

Step 14:- Update anyone with marks less than 65 to have a "fail" status using follwoing command

```
db.students.updateMany({marks: {$lt: 65}}, {$set: {status: "fail"}})
```

```
studentDB> db.students.updateMany({marks: {$lt: 65}}, {$set: {status: "fail"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

Step 15:- Delete Rutvik (Roll No 1) from the database using the following command

```
db.students.deleteOne({rollno: 1})
```

```
studentDB> db.students.deleteOne({rollno: 1})
{ acknowledged: true, deletedCount: 1 }
```

Step 16:-If instead of printing out all the data, if we just want exactly how many students are left in the database then use the following command.

```
db.students.countDocuments()
```

```
studentDB> db.students.countDocuments()
4
```

Step 17:-Print all the records

```
db.students.find().pretty()
```

```
[
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b7'),
    rollno: 2,
    name: 'Aditya',
    marks: 85,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b8'),
    rollno: 3,
    name: 'Priya',
    marks: 80,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b9'),
    rollno: 4,
    name: 'Amit',
    marks: 75,
    city: 'Mumbai'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8ba'),
    rollno: 5,
    name: 'Sneha',
    marks: 82,
    status: 'pass',
    city: 'Mumbai'
  }
]
```

Step 18:-Now stop Mongo DB service to prevent your laptop/computer from draining power or if you want to exit from Mongo DB shell then use the following command:-

```
sudo service mongod stop
```